

CPE 381 CLASSWORK 06: DISCRETE DIFFERENCE, OPERATIONS ON SIGNALS

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 14, 2026 (In-Class)

1. Taylor series expansion around operating point t_0 is given by

$$x(t_1) = x(t_0 + h) = x(t_0) + hx'(t_0) + \frac{1}{2!}h^2x''(t_0) + \cdots +$$

$$x(t_1) \approx x(t_0) + hx'(t_0)$$

where $t_1 = t_0 + h$.

Consider $x(t) = \cos(t)$, and assuming $h = 0.1$, and $t_0 = 0$, what is $\cos(t_1)$? **(5 Points)**

Answer

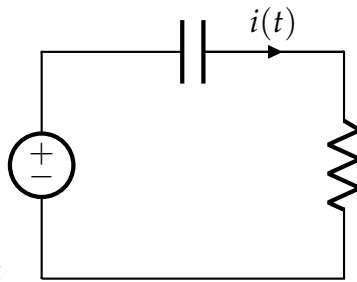
$$x'(t) = -\sin(t)$$

$$\cos(t_1) \approx \cos(t_0) + h(-\sin(t_0))$$

$$\approx \cos(0) + 0.1(-\sin(0))$$

$$\cos(1.1) \approx 1$$

Note that if we add next term in Taylor series, approximation will change and move closer to the true value.



2. Consider a circuit that has a constant voltage source $v_i(t)$ with $v_c(t)$ as the voltage across the capacitor. Its differential equation is given by

$$v_i(t) = v_c(t) + \frac{dv_c(t)}{dt}$$

Assume $v_c(t) = 0$ for $t \leq 0$. Further, assume Δt as the constant sampling time and write down the expression for $v_i(t)$ in the form of discrete difference equation by considering two-point backward difference as the form to compute discrete derivative. **(10 Points)**

Answer

In two-point backward difference:

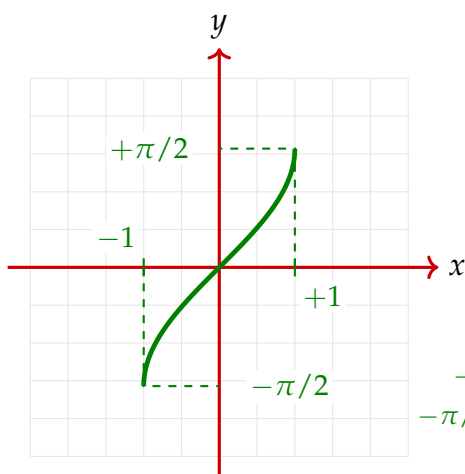
$$\frac{dv_c(t)}{dt} = \frac{v_c(t) - v_c(t - \Delta t)}{\Delta t}$$

Thus,

$$v_i(t) = v_c(t) + \frac{v_c(t) - v_c(t - \Delta t)}{\Delta t}$$

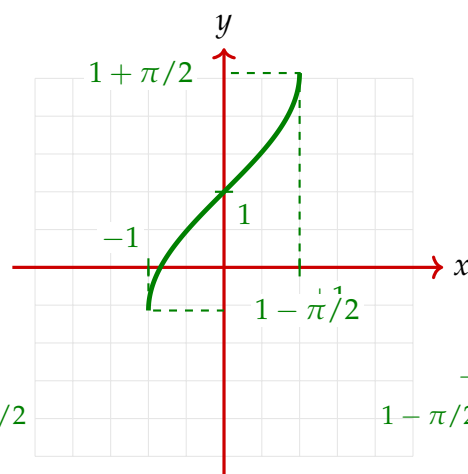
3. $y = \sin^{-1}(x)$ is given. Plot $y_1 = \sin^{-1}(x) + 1$ on the empty grid. **(2.5 Points)**

$$y = \sin^{-1}(x)$$



$$\begin{aligned} -1 \leq x \leq 1 \\ -\pi/2 \leq y \leq \pi/2 \end{aligned}$$

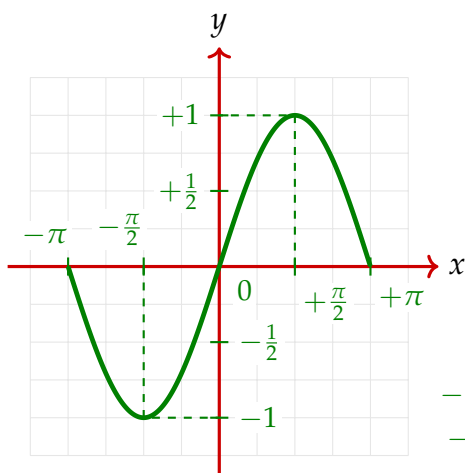
$$\text{Solution: } y_1 = \sin^{-1}(x) + 1$$



$$\begin{aligned} -1 \leq x \leq 1 \\ 1 - \pi/2 \leq y_1 \leq 1 + \pi/2 \end{aligned}$$

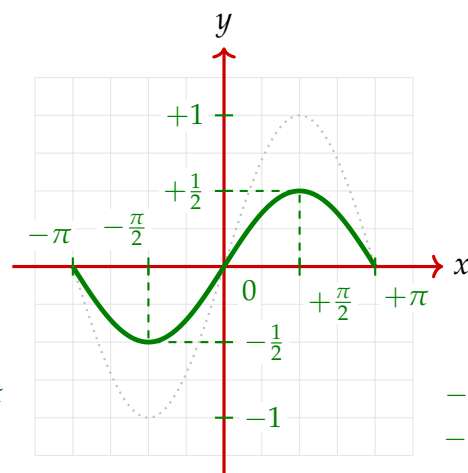
4. Consider $y = \sin(x)$. Based on this, plot $y_1 = \frac{1}{2} \sin(x)$. **(2.5 Points)**

$$y = \sin(x)$$



$$\begin{aligned} -\pi \leq x \leq \pi \\ -1 \leq y \leq 1 \end{aligned}$$

$$y_1 = \frac{1}{2} \sin(x)$$



$$\begin{aligned} -\pi \leq x \leq \pi \\ -\frac{1}{2} \leq y_1 \leq \frac{1}{2} \end{aligned}$$